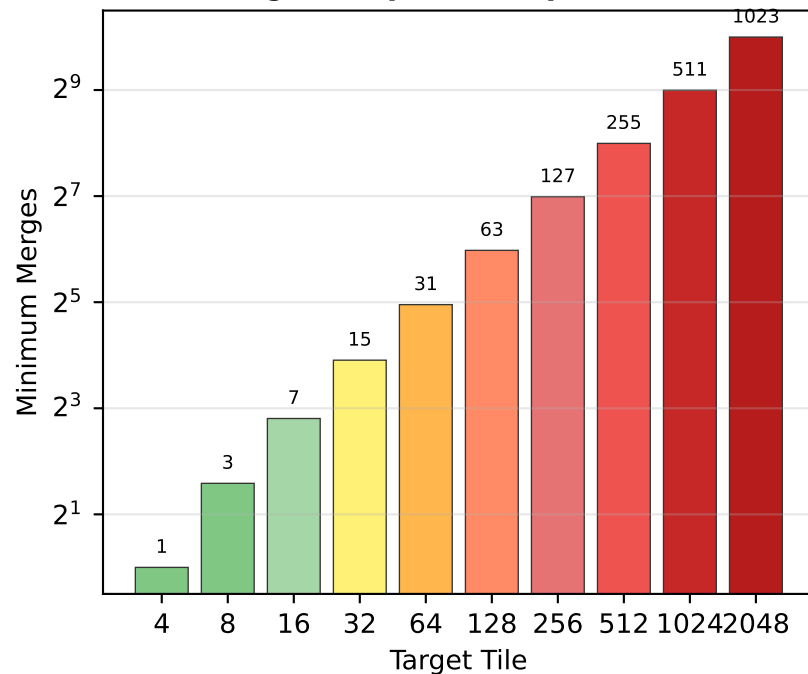
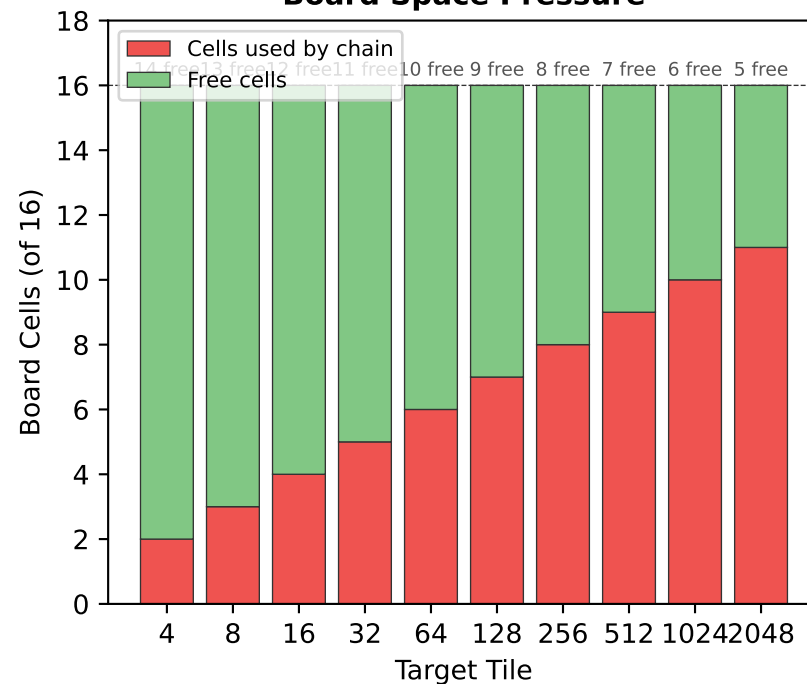


2048 Tile Difficulty Analysis

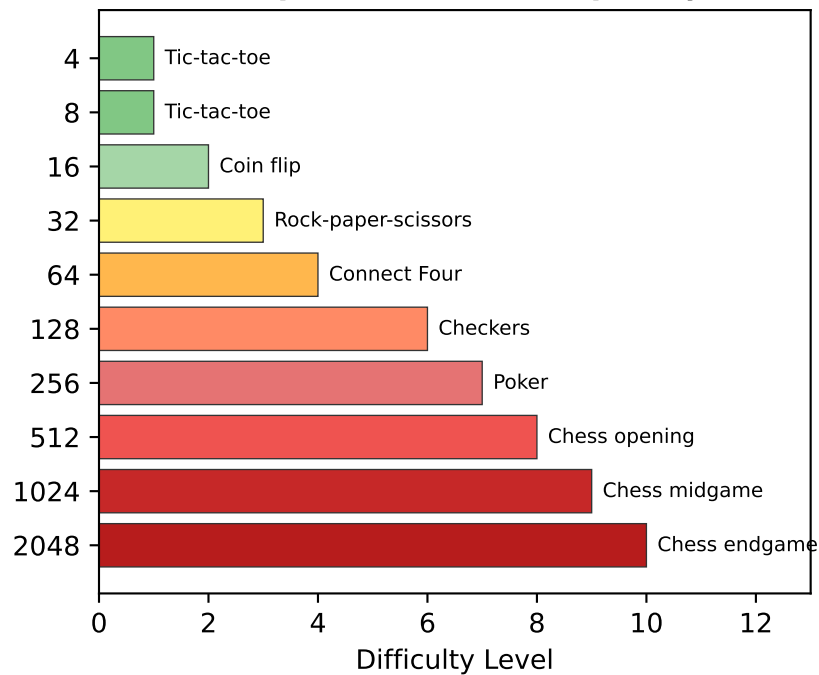
Merges Required (exponential)



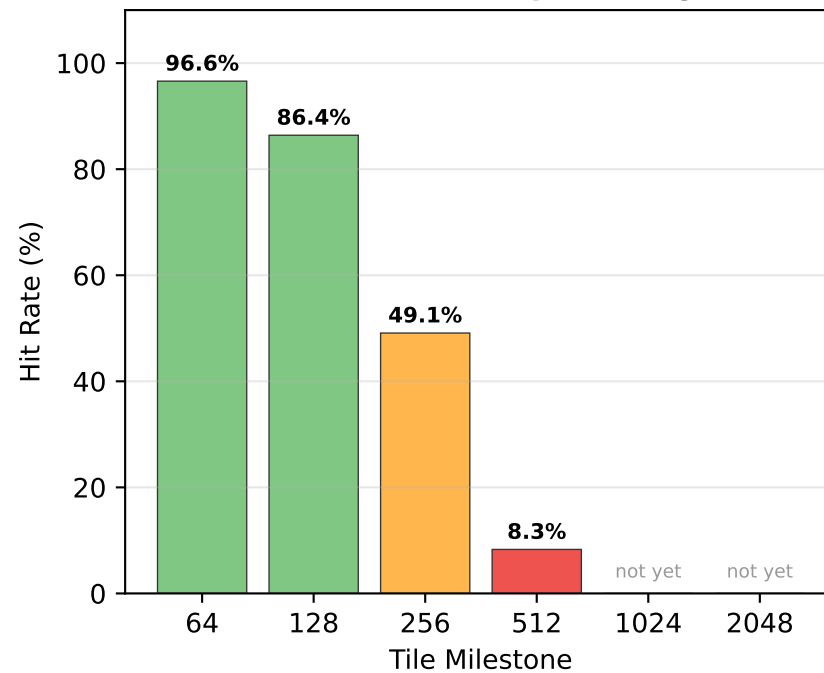
Board Space Pressure



Comparable Game Complexity



GENREG Hit Rates (868 params, gen 63K)



Why Each Tile Is Exponentially Harder

Tile	Merges	Chain Cells	Free Cells	Human Analogy	What This Actually Means
4	1	2	14	Stacking 2 blocks	Two 2-tiles appear next to each other. Push them together.
8	3	3	13	Like tic-tac-toe: just match pairs	Merge 2->4, then merge two 4s. A toddler can do this.
16	7	4	12	Random button mashing gets here ~60% of the time	No strategy needed. Random moves produce enough accidental merges. This is the "noise floor."
32	15	5	11	Learning to always push one direction first	The agent must learn directional preference. This is where evolution starts to matter.
64	31	6	10	Like Connect Four: plan 2-3 moves ahead	Must keep big tiles together. Scattering tiles across the board is now fatal. Requires spatial awareness.
128	63	7	9	Like checkers: manage multiple pieces at once. 7 tiles in a specific chain.	Must anchor the largest tile in a corner and build a descending chain. Requires multi-step planning and the ability to recover from disruption.
256	127	8	8	Like poker: read the board, manage risk. Half your board is committed.	Managing 8 tile levels simultaneously. The board is half-full just from the chain. Every merge must be sequenced correctly or the chain breaks.
512	255	9	7	Like a chess opening: every move matters. One mistake kills the run.	The margin for error is near zero. The agent must plan 10+ moves ahead, maintain a 9-level chain, and avoid trapping tiles. Like landing a plane.
1024	511	10	6	Like chess midgame: 500+ perfect moves. 10-tile chain, 6 cells to work in.	Requires near-perfect play for 500+ consecutive decisions. One suboptimal move cascades into death. Fewer than 1% of human players reach this.
2048	1023	11	5	Like solving a Rubik's cube blindfold: 1000+ perfect decisions, 5 free cells.	Theoretical perfection over 1000+ moves. Only achievable with deep lookahead or superhuman pattern recognition. The "moon landing" of 2048.

Each tile doubles the required merges. The board has only 16 cells — reaching higher tiles means less room to maneuver. GENREG achieves 512-tile play with 868 parameters and zero game-specific reward engineering.